For instance, imagine a chat application where the client stores the last 10 messages for quick display. The server, however, maintains the entire conversation history for context and security. When the user sends a new message, the client adds it to its local state and sends it to the server. The server then processes this message, potentially using it with the full history to generate an AI response. The server's response is then sent back to the client, which updates its local state. This process ensures that the client has fast access to recent messages while the server maintains a secure, centralized record of the entire conversation.

But state management is just the tip of the iceberg. As user expectations and business requirements escalate, new challenges emerge. How do we instruct AI models to generate structured data, such as JSON, instead of raw text? Once generated, how can we validate this data to ensure its correctness and usability? Asking those questions early helps reveal the growing complexity of AI-powered web applications and the need for advanced solutions.

React server components (RSCs) and the Vercel AI SDK can provide solutions to some of these concerns by improving the way you build and manage state in AI applications, significantly enhancing performance and user experience. You also need to know how to generate structured, type-safe data directly from AI models, a capability that bridges the gap between raw AI outputs and the structured data your application needs. This approach not only improves reliability but also opens new possibilities for integrating AI-generated content effortlessly into various tools and services.

Finally, *tool* and *function calling* in AI models can greatly expand the capabilities of AI applications. By enabling AI to interact with external resources and perform complex operations, you can create more flexible and stronger web applications that can adapt to real-time data and user needs. Figure 4.1 shows how all of these pieces interconnect.

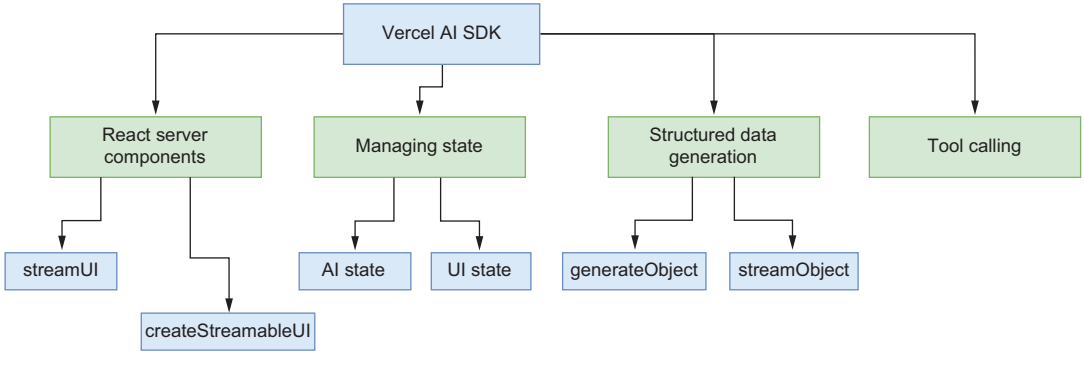


Figure 4.1 The four main topics of this chapter

Now let's update our Astra AI project to incorporate these advanced techniques.